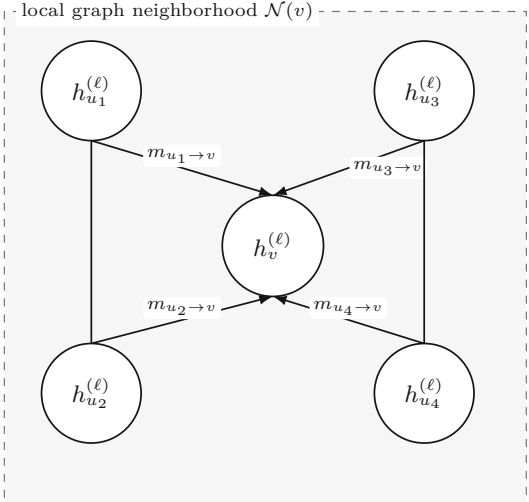
Graph message passing

Conceptual one-hop aggregation followed by a node-state update



Message on each edge

$$m_{u \to v} = M_\ell(h_v^{(\ell)}, h_u^{(\ell)}, e_{uv})$$

Permutation-invariant aggregation

$$m_v = \sum_{u \in \mathcal{N}(v)} m_{u \to v}$$

Node update

$$h_v^{(\ell+1)} = U_\ell(h_v^{(\ell)}, m_v)$$

e_{uv} is an optional edge feature.
Graph size and hidden width are unspecified.

Conceptual MPNN update after Gilmer et al. (2017); the same local rule is applied across graph nodes.